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Chapter 1

Introduction

Chapter 2

Architectural Design

2.1 Overview

The **Best Bike Paths (BBP)** system is architected as a **Hybrid Mobile Application** supported by a **Serverless Backend-as-a-Service (BaaS)** infrastructure.

This design strategy allows for a unified codebase (using web technologies like TypeScript/React) to be deployed across both Web and Mobile platforms. Crucially, by leveraging a native runtime bridge (e.g., Capacitor), the application overcomes the limitations of standard web apps. This bridge grants access to native device features, specifically background geolocation and high-frequency motion sensors, which are required for the road Quality Detection and reliability requirements defined in the RASD.

Unlike traditional N-Tier architectures, BBP leverages a thick client model. The client applications contain a significant amount of application logic, communicating directly with the managed cloud database via secure APIs. Complex business logic requiring data aggregation or external integrations is offloaded to edge functions, which execute on-demand in the cloud.

2.1.1 High-Level Architecture Description

The system is composed of two main logical nodes:

1. **Hybrid Client Tier:** This tier includes the unified application running on user devices.
    - **Mobile Deployment (Android/iOS):** The application runs within a native container using Capacitor. It accesses hardware sensors via native plugins to perform precise activity tracking and road irregularity detection even when the device screen is off.
    - **Web Deployment:** The same codebase serves as a standard Progressiva Web App for path planning and profile management.
    - **Responsibilities:**
        - Acquisition and pre-processing of raw sensor data (GPS, Accelerometer) via the Native Bridge.
        - Local data caching (Offline-First) to ensure reliability.
        - Direct communication with the BaaS provider for CRUD operations.
2. **Serverless Cloud Tier:** Hosted on a BaaS platform (e.g., Supabase), acting as the scalable backend. It provides:
    - **Data Persistence:** A PostgreSQL database extended with PostGIS for spatial operations.
    - **Authentication:** A built-in identity provider handling OAuth2 and JWT generation.
    - **Edge Logic:** Stateless serverless functions that handle background tasks such as path merging, ranking updates, and weather data enrichment.

2.1.2 Security Model: Row Level Security (RLS)

Since clients communicate directly with the database, security is enforced at the database layer using Row Level Security (RLS) policies. Upon login, the Auth Service issues a JSON Web Token (JWT) to the client, which is included in every database query. The database engine evaluates RLS policies for each row to determine if the operation is permitted. This ensures that "direct access" does not imply "unrestricted access," maintaining strict data isolation and integrity.

2.2 Component view

This section details the software components within the Client and Cloud tiers, defining their specific responsibilities and interactions.

2.2.1 Hybrid Client Components

- **Native Bridge (Capacitor):** The architectural boundary between the web code and the native OS. It exposes a unified API to access hardware features. It ensures that the **Geolocation** and **Accelerometer** continue to operate reliably even when the application is in the background, bridging the gap between the web runtime and the device hardware.
- **Sensor Controller & Filter:** Responsible for logic related to data acquisition.
  - It interfaces with the **Native Bridge** to subscribe to GPS and Accelerometer streams.
  - It implements a **Kalman Filter** to smooth raw GPS coordinates.
  - It analyzes accelerometer data to detect vertical shocks (road surface irregularities like potholes) and buffers this data locally during activity recording.
- **Local Persistence Manager:** Implements the "Offline-First" capability. It uses an embedded database (e.g., SQLite via Capacitor plugin) to store activity traces and pending reports when the network is unavailable. It includes a **Sync Engine** that monitors network reachability and uploads buffered data to the cloud once connectivity is restored.
- **BaaS Client SDK:** The abstraction layer provided by the BaaS platform. It manages the persistent WebSocket connection for real-time updates, handles secure API calls to the database, and automatically refreshes authentication tokens.
- **Map & UI Controller:** Manages the visualization of map tiles and path geometries. It handles user inputs (forms, search queries) and renders the visual feedback for the navigation system.

2.2.2 Managed Cloud Components (BaaS)

- **Auth Service (Identity Provider):** Manages user lifecycles (sign-up, login, password recovery). It integrates with third-party OAuth providers (e.g., Google, Facebook) and issues secure JWTs containing user roles and IDs.
- **Spatial Database (PostgreSQL + PostGIS):** The central data repository. It is an active component that:
  - Enforces **RLS Policies** to secure data access.
  - Executes complex **Spatial Queries** (e.g., `ST_DWithin`, `ST_Intersects`) natively, allowing clients to efficiently retrieve paths within a specific viewport.
  - Triggers database events (Webhooks) to activate Edge Functions upon data changes.

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- **Object Storage:** A specialized file storage system for large binary assets, such as user profile pictures or images attached to path reports.

2.2.3 Edge Functions

These are stateless functions triggered by database events or scheduled cron jobs:

- `func_path_merging`
  - **Trigger:** Database event (INSERT on `Activities` or `Feedbacks` table).
  - **Responsibility:** It implements the **Data Consolidation & Consensus Logic**. When new data (an activity trace or a manual report) is associated with a path, this function aggregates it with existing reports to determine the path's `CurrentCondition`. It applies a **Majority Voting Algorithm** weighted by data freshness (e.g., resolving a conflict where 3 recent reports say "Good" and 1 says "Bad" by marking the segment as "Good") to ensure the status reflects the actual road situation.
- `func_path_search`
  - **Trigger:** Direct Client Request.
  - **Responsibility:** It implements the **Search & Ranking Algorithm** defined in the RASD. When a user queries for a route (specifying Origin and Destination), this function:
    - Performs a spatial query to find candidate paths connecting the points.
    - Dynamically calculates a **Suitability Score** for each candidate. This score combines the **Geospatial match** (how well it fits the requested Start/End points) with the **Path Quality** (derived from the consolidated feedback and applying a time-decay factor to older data).
    - Returns the list of paths ordered by this score.
- `func_maintenance_job`
  - **Trigger:** Scheduled Cron Job (e.g., Nightly).
  - **Responsibility:** Periodically applies the **Time-Decay Function** to the database. It archives or lowers the weight of very old reports/feedbacks to ensure that the `func_path_search` always prioritizes fresh information without needing to process the entire history during every query.
- `func_weather_enricher`
  - **Trigger:** Database event (INSERT on `Activities`).
  - **Responsibility:** It securely calls external Meteorological APIs (using backend-only API keys) to fetch weather conditions for the specific time and location of the activity, enriching the path metadata.

2.3 Deployment view

The deployment architecture of **Best Bike Paths** follows a distributed, client-server model relying on a **Serverless Infrastructure**. This approach minimizes operational overhead, as the backend resources (Database, Auth, Compute) are provisioned and scaled automatically by the Cloud Provider.

The system is deployed across three primary types of nodes:

1. **Mobile User Device (Android/iOS):**

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- **Hardware:** Personal Smartphone.
- **Execution Environment:** Native OS hosting the **Capacitor Runtime** (WebView).
- **Artifacts:** The *Hybrid Mobile Application* package (`.apk` / `.ipa`). This artifact contains the compiled Javascript/React bundle and the native plugins required to access hardware sensors (GPS, Accelerometer). It includes an embedded SQLite database for local persistence.

2. **Web User Station (Desktop/Laptop):**
   - **Hardware:** Personal Computer.
   - **Execution Environment:** Modern Web Browser (Chrome, Firefox, Safari).
   - **Artifacts:** The *Progressive Web Application (PWA)* assets (HTML, CSS, JS bundles) cached by the Service Worker.
3. **Managed Cloud Platform (BaaS Provider):**
   - **Hardware:** Distributed Virtualized Clusters (managed by Supabase/AWS).
   - **Execution Environment:** Containerized Database Engine and V8 Isolate runtime for Edge Functions.
   - **Artifacts:**
     - *PostgreSQL Instance:* Stores data and RLS policies.
     - *Auth Service Instance:* Handles identity management.
     - *Edge Functions:* Deployed serverless scripts (`path_merging`, `score_calc`) executed on demand.

2.3.1 Network Communication

Communication between the Client Nodes and the Cloud Platform occurs exclusively over secure standard protocols:

- **HTTPS (TLS 1.3):** For all REST API calls (Database CRUD, RPC functions).
- **WSS (Secure WebSockets):** For real-time event subscriptions (e.g., live feedback updates).

2.4 Runtime view

2.5 Component interfaces

This section specifies the APIs provided and required by the major components identified in the Component View. The interfaces are categorized into **Hardware Abstraction Interfaces** (internal to the Hybrid Client) and **Network Interfaces** (between Client and Cloud).

2.5.1 Hardware Abstraction Interfaces

These interfaces are exposed by the **Native Bridge (Capacitor)** and consumed by the **Sensor Controller** component within the web runtime. They allow the TypeScript application logic to access device sensors.

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Interface: IGeolocationProvider

- **Description:** Provides access to the device's GPS hardware.
- **Provided By:** Native Bridge (`@capacitor/geolocation`)
- **Used By:** Sensor Controller

Method Signature	Input Parameters	Return Value	Description
`getCurrentPosition()`	`options: GeoOptions`	`Promise<Position>`	Returns the single current coordinate (Lat, Long, Altitude).
`watchPosition()`	`options: GeoOptions, callback: (pos) => void`	`CallbackID`	Starts a continuous background location tracking session. The callback is invoked every time the OS receives a location update.
`clearWatch()`	`id: CallbackID`	`void`	Stops the tracking session to save battery.

Table 2.1: IGeolocationProvider Interface

Interface: IMotionSensor

- **Description:** Provides high-frequency access to the accelerometer for pothole detection.
- **Provided By:** Native Bridge (`@capacitor/motion`)
- **Used By:** Sensor Controller (Inertial Analyzer)

Method Signature	Input Parameters	Return Value	Description
`addListener()`	`eventName: 'accel', callback: (event) => void`	`PluginListenerHandle`	Subscribes to the raw accelerometer stream. The event contains x, y, z acceleration vectors used to detect vertical shocks.
`removeAllListeners()`	`void`	`void`	Unsubscribes from the sensor stream.

Table 2.2: IMotionSensor Interface

2.5.2 Cloud Network Interfaces (BaaS API)

These interfaces represent the communication contract between the **Hybrid Client** and the **Serverless Cloud**. Since the architecture uses a BaaS, these are primarily RESTful endpoints automatically generated by Supabase from the database schema, plus RPC (Remote Procedure Calls) for Edge Functions.

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Interface: IAuthService

- **Description:** Manages user sessions.
- **Protocol:** HTTPS / REST
- **Provided By:** Auth Service (Supabase GoTrue)

Endpoint / Method	Input Payload	Response	Description
`POST /auth/v1/signup`	`{ email, password }`	`{ user, session, jwt }`	Registers a new user and returns the access token.
`POST /auth/v1/token`	`{ email, password, grant_type }`	`{ access_token, refresh_token }`	Authenticates a user (Login).
`POST /auth/v1/logout`	Header: `Bearer <token>`	`204 No Content`	Invalidates the current session.

Table 2.3: IAuthService Interface

Interface: ISpatialDataStore

- **Description:** Provides CRUD access to geospatial entities. Access is filtered by RLS policies.
- **Protocol:** HTTPS / REST
- **Provided By:** Spatial Database (PostgREST)

Endpoint / Method	Input Payload	Response	Description
`GET /rest/v1/paths`	Query Params: `select=*, geometry=...`	`JSON Array`	Retrieves public paths. Supports spatial filtering (e.g., retrieving paths within a map viewport).
`POST /rest/v1/activities`	`{ user_id, geometry_linestring, ... }`	`201 Created`	Uploads a recorded activity trace. This action triggers the `func_path_merging` Edge Function asynchronously.
`POST /rest/v1/feedbacks`	`{ path_id, rating, comment }`	`201 Created`	Submits a user report for a specific path.

Table 2.4: ISpatialDataStore Interface

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Interface: ISearchEngine (RPC)

- **Description:** Exposes the complex path ranking logic as a remote function call.
- **Protocol:** HTTPS / RPC
- **Provided By:** Edge Functions

Function Name	Input Parameters	Return Value	Description
`rpc/search_best_paths`	`{ origin_lat, origin_lon, dest_lat, dest_lon }`	`JSON Array <Path>`	Invokes the routing algorithm. Returns a list of paths ordered by the dynamic **Suitability Score** (calculated based on distance and road quality).

Table 2.5: ISearchEngine Interface

2.5.3 External Interfaces

Interface: IWeatherProvider

- **Description:** External API used to enrich path data with meteorological conditions.
- **Provided By:** Third-party provider (e.g., OpenWeatherMap)
- **Used By:** `func_weather_enricher` (Edge Function)

Endpoint	Method	Params	Description
`/data/2.5/weather`	`GET`	`lat, lon, appid`	Fetches current weather data (Temperature, Rain volume, Wind) for a specific coordinate.

Table 2.6: IWeatherProvider Interface

2.6 Selected architectural styles and patterns

2.6.1 Architectural Styles

- **Serverless / Backend-as-a-Service (BaaS)**
  - **Description:** The system does not rely on a traditional, self-managed monolithic application server. Instead, backend responsibilities, including database management, authentication, and file storage, are delegated to a managed cloud provider (Supabase). Application logic is executed via stateless containers (Edge Functions) that spin up only when triggered.
  - **Rationale:** This style was chosen to maximize scalability and minimize operational overhead. Since the "Best Bike Paths" system must handle potential spikes in usage without manual provisioning, a serverless infrastructure ensures automatic scaling. Furthermore, it allows the development team to focus entirely on the complex domain logic (merging and ranking) rather than server maintenance and patching.

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- **Event-Driven Architecture**
  - **Description:** The interaction between data ingestion and data processing is decoupled. When a client uploads a new activity, the system does not process it synchronously. Instead, the database emits an event (webhook), which asynchronously triggers the specific Edge Function responsible for the merging algorithm.
  - **Rationale:** The Path Merging process involves complex geospatial calculations (intersections, buffering) that are computationally expensive. Processing them synchronously would block the client UI and lead to connection timeouts. An event-driven approach ensures the user interface remains responsive (Latency Hiding) while heavy tasks are processed in the background.
- **Thick Client**
  - **Description:** A significant portion of the application logic resides on the client device rather than the server. The Hybrid Client is responsible for sensor data acquisition, noise filtering (Kalman Filter), data validation, and local buffering.
  - **Rationale:** This style is strictly required to satisfy the reliability constraints defined in the RASD. Cyclists often traverse areas with no network coverage; the client must be able to function autonomously, recording and validating data locally, and syncing only when connectivity is restored.

2.6.2 Design Patterns

- **Bridge Pattern (via Capacitor)**
  - **Description:** The application utilizes a native bridge to decouple the high-level application logic (written in TypeScript/React) from the low-level platform implementation (Android/iOS). The web code calls a uniform JavaScript API, and the Bridge translates these calls into native OS commands.
  - **Rationale:** This pattern enables cross-platform accessibility while satisfying the need for hardware-specific features. Specifically, Capacitor native plugins allows the web-based application to access the accelerometer and background geolocation services, which are otherwise inaccessible or limited in a standard browser environment.
- **Observer Pattern**
  - **Description:** The client application subscribes to real-time changes on specific database tables (e.g., `Feedbacks`) using Secure WebSockets (WSS). When the data changes on the server, the new state is automatically pushed to all listening clients.
  - **Rationale:** This is essential for the safety-critical feature of reporting road hazards. If a user reports a danger, other nearby cyclists must be notified immediately to update their map view, without needing to manually refresh the application.
- **Row Level Security (RLS)**
  - **Description:** Access control logic is implemented directly within the database engine rather than in the application code. Each database query is evaluated against a set of SQL policies that check the user's authentication token against the data being accessed.
  - **Rationale:** Since the architecture allows clients to communicate directly with the database (skipping the middleware layer), RLS is the only mechanism to guarantee dData isolation. It mathematically ensures that a user can never read or modify another user's private data, regardless of any potential bug in the client code.

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2.7 Other design decisions

2.7.1 Adoption of Hybrid Mobile Architecture

- **Description:** The development strategy involves building a single web-based codebase wrapped in a native container (Capacitor), rather than developing two separate native applications (Kotlin for Android, Swift for iOS) or a pure Progressive Web App (PWA).
- **Rationale:** A pure PWA was discarded because it cannot reliably access high-frequency motion sensors in the background, which is critical for the "Road Quality Detection" requirement. Native development was discarded due to the high effort required for a small team. The Hybrid approach offers the necessary hardware access with a unified codebase, significantly reducing development and testing time.

2.7.2 Usage of PostgreSQL with PostGIS

- **Description:** The system utilizes a Relational Database Management System (RDBMS) extended with the PostGIS spatial engine to store and query all geospatial data (Points, Linestrings, Polygons).
- **Rationale:** Generic databases are insufficient for the mathematical complexity of this project. PostGIS provides native, highly optimized functions for geospatial conflation and Nearest-Neighbor searches. Implementing these geometric algorithms from scratch in code would be error-prone and inefficient.

2.7.3 Centralized Logic for Merging and Ranking

- **Description:** The core algorithms that determine the shape of public paths (Merging) and their value (Ranking) are executed exclusively on the Cloud Tier via Edge Functions, never on the client.
- **Rationale:** This decision ensures a **Single Source of Truth**. The merging algorithm requires a complete view of all existing paths in an area to make a decision, which a single client does not possess. Furthermore, calculating the Ranking on the server prevents malicious users from tampering with the client code to artificially inflate the score of their preferred paths.

2.7.4 Direct-to-Database Communication Pattern

- **Description:** The architecture bypasses the traditional "API Gateway / Controller" layer for standard operations. Clients perform CRUD (Create, Read, Update, Delete) operations directly against the database's REST API.
- **Rationale:** The majority of user interactions (e.g., loading a profile, saving a track) are standard data operations that do not require complex business logic. Implementing a dedicated middle-layer for these simple tasks would introduce unnecessary latency and code maintenance ("Boilerplate"). Security is delegated to RLS (as described in 2.6.2), making the middle layer redundant for these tasks.

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Chapter 3

User Interface Design

Chapter 4

Requirements Traceability

Chapter 5

Implementation, Integration and Test Plan

Chapter 6

Effort Spent

Chapter	Date	Alessio P.	Alberto S.	Luca S.
Introduction	29/12/2025	1h	1h	-
	Total			
Architecture Design	30/12/2025	-	3.5h	-
	31/12/2025	-	2h	-
	01/01/2026	-	2h	-
	Total			
User Interface Design	date	1h	1h	1h
	Total			
Requirements Traceability	date	1h	1h	1h
	Total			
Implementation, Integration and Test Plan	date	1h	1h	1h
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